

## **Weekly Report - 5/11/2025**

Group1:(Anik Barua, A.Alim, Farea Mahdea)

### **1. Tasks Completed:**

- **Anik's Paper Reading:**

- This week, Anik read and analyzed three related papers on sleep stage classification and anomaly detection. Key insights were extracted and noted, which will be helpful for the ongoing project on context-aware sleep stage classification.

- **A. Alim's Work:**

- A. Alim read two related papers on hybrid models for time-series anomaly detection and provided feedback. The findings from these papers were incorporated into the project's approach, specifically focusing on model hybridization (CNN-LSTM-Transformer).
- PDFs of the two papers read by A. Alim were organized and stored for future reference.

- **Work Plan for FareaMahdea:**

- A detailed work plan was prepared for FareaMahdea, outlining tasks related to the sleep disorder classification project.
- Key objectives include data preprocessing, feature extraction, and setting up the baseline CNN-LSTM model.
- A clear timeline was created to ensure milestones are met on time for this phase of the project.

- **Paper Reading & Proposed Report:**

- FareaMahdea reviewed relevant papers and proposed solutions related to personalized anomaly detection and attention mechanisms. The insights were used to inform the current project report, with special focus on the hybrid architecture and its benefits.
- The final proposed report was compiled, integrating all the learnings and strategies discussed this week.

- **Combining Decisions and Fixes:**

- All discussions were consolidated, and final decisions were made on the choice of models (CNN-LSTM-Transformer hybrid) and methods (personalized anomaly scoring, attention visualization).
- Necessary fixes were applied to the project outline and strategy to ensure smoother implementation and clearer goals moving forward.

## **2. Summary of Key Progress:**

- Anik's reading has expanded the knowledge base for the project, particularly regarding sleep stage classification.
- A. Alim's input helped refine the hybrid model architecture, integrating concepts from related papers.
- FareaMahdea's feedback on the report has ensured that the proposed methodology aligns with the project objectives.
- The team has a clearer roadmap for next steps and is ready to move into the next phase of implementation.

## **3. Next Steps for Next Week:**

- Continue with data collection and preprocessing tasks as outlined in the work plan.
- Begin the implementation of the baseline CNN-LSTM model for sleep stage classification.
- Review and refine the hybrid model architecture based on the literature readings and feedback.
- Start training the hybrid model and evaluating its performance.

## **4. Challenges Faced:**

- No major challenges were encountered this week, though further adjustments might be required during the model implementation phase.